

FinTrackPro: Real Application Screenshots & Visual Manual

Production Interface Walkthrough, Relational Schemas & System Modules

S.I.E.S College of Arts, Science and Commerce, Sion(W), Mumbai – 400 022

Department of Computer Science • University of Mumbai (Academic Year 2025–2026)

PROJECT & CANDIDATE SPECIFICATION

Project Title:	FinTrackPro (WealthFlow / Velocity Wealth Platform)
Submitted By:	PARTHASARATHY RADHAKRISHNAN
Roll Number:	TCS2627061
Project Guide:	Prof. Maya Nair, Assistant Professor
Head of Department:	Dr. Manoj Singh, Head of Computer Science

Included Screens: Investor Dashboard, Fund Screener, Advisor CRM, KYC Vault, NACH E-Mandate, DB Schemas

CORE MODULE: Consolidated Portfolio Analytics, Multi-AMC Folio Tracking & Newton-Raphson XIRR

Real-time valuation across equity, debt, and gold schemes with daily AMFI NAV updates, SIP status, and asset allocation breakdown.

Screen Component	Live Metric / Visual Display	Mathematical / Business Logic	Status
Total Portfolio Value	¹ 18,45,280	Sum of (Current NAV × Units Held) across all AMC folios	LIVE
Annualized Yield	+ 15.42% XIRR	Newton-Raphson polynomial solver across irregular monthly SIP dates	VERIFIED
Unrealized Gain	+ ¹ 4,25,280 (+ 29.95%)	Net Capital Gains (Current Valuation - Cumulative Cost Basis)	REAL-TIME
Asset Donut Chart	Equity 65.2% Debt 25.1% Gold 9.7%	Recharts composable SVG donut visualizing target vs actual asset drift	ACTIVE
Active SIP Commitments	¹ 35,000 / month across 12 sch	Next automated bank debit on 10th Oct 2026 via NPCI NACH	HEALTHY

CORE MODULE: High-Throughput Scheme Screener & Live NAV Catalog

Direct search across all Indian AMCs (HDFC, ICICI, SBI, Quant, Nippon, Parag Parikh) with live NAVs from official AMFI feeds.

Scheme Code	Scheme Name	Category	Latest NAV (₹)	3Y Return (CAGR)	AUM (Cr)
120823	Motilal Oswal Midcap Fund - Direct Plan - Growth	Equity - Mid Cap	₹ 98.45	+ 32.40%	₹ 16,420 Cr
118556	Tata Digital India Fund - Direct Plan - Growth	Sectoral - Tech	₹ 48.12	+ 24.18%	₹ 9,850 Cr
122639	Parag Parikh Flexi Cap Fund - Direct - Growth	Equity - Flexi Cap	₹ 84.92	+ 21.85%	₹ 68,140 Cr
120465	Nippon India Growth Fund - Direct Plan - Growth	Equity - Mid Cap	₹ 3,842.10	+ 28.92%	₹ 27,900 Cr
108466	DSP Small Cap Fund - Direct Plan - Growth	Equity - Small Cap	₹ 162.80	+ 26.40%	₹ 14,200 Cr
120700	Canara Robeco Emerging Equities - Direct Plan	Large & Mid Cap	₹ 224.50	+ 19.80%	₹ 21,300 Cr

CORE MODULE: Velocity Wealth AMFI Partner Portal (ARN: 348996)

Institutional multi-tenant dashboard allowing advisors to onboard investors, verify KYC, track AUM, and audit brokerage.

Client Name	PAN Number	Total AUM	Monthly SIP Book	KYC Status	Advisor Action
Rajesh Mehra	ABCDE1234F	₹ 28,45,000	₹ 40,000 / mo	VERIFIED	Active (Rebalance Review)
Sneha Deshmukh	BNMPK5678Q	₹ 14,80,200	₹ 25,000 / mo	SUBMITTED	Pending KYC Document Approval
Amitabh Sen	CZXPS9821L	₹ 42,10,000	₹ 55,000 / mo	VERIFIED	Active (Annual Review Due)
Vikramaditya Iyer	DFGPS3412M	₹ 8,90,000	₹ 15,000 / mo	VERIFIED	Active (SIP Added)
Pooja Kulkarni	JKLMN9012R	₹ 6,50,000	₹ 10,000 / mo	VERIFIED	Active

CORE MODULE: Paperless e-KYC Onboarding & Biometric / Signature Vault

SEBI and KRA compliant identity verification with live PAN regex check, masked Aadhaar UIDAI integration, and HTML5 signature pad.

Verification Step	Security Standard / Protocol	Cryptographic Check	Regulatory Compliance
PAN Validation	Algorithmic 4th char entity check (P = Individual)	Synchronous NSDL checksum verification	SEBI Master Circular 2023
Aadhaar Masking	Strict 8-digit masking (UIDAI Circular 2021)	SHA-256 hash storage, no raw UID stored	Aadhaar Act Section 29
Digital Signature Pad	HTML5 Canvas capturing path vectors at 60 FPS	Base64 PNG encoded with tamper-evident salt	Information Technology Act 2000
Bank Account Penny Drop	NPCI IMPS verification of IFSC & Account Name	100% name match against PAN database	RBI Fraud Prevention Guidelines
Document Audit Vault	AES-256 encrypted storage in Cloud SQL / Firestore	Immutable append-only audit trail	DPDP Act 2023 Compliance

CORE MODULE: NPCI National Automated Clearing House (NACH) 3D Secure Registration

Enables recurring bank auto-debits with user-specified debit days, max daily mandate caps, and instant Net Banking e-Sign.

Mandate Parameter	Client Configuration	NPCI Settlement Specification	Failure Fallback Mode
Sponsor Bank	HDFC Bank Ltd (IFSC: HDFC0000128)	Direct NPCI Gateway Router	Retry via Secondary Bank
Mandate Limit	¹ 50,000 / day maximum debit c	Customer protection guard against excess debit	Auto-Reject on Limit Breach
SIP Frequency	Monthly (10th of every calendar month)	T-2 bank notification dispatched via SMS	Next business day on holidays
Authentication	Net Banking 3D Secure / Debit Card OTP	NPCI UMRN (Unique Mandate Reference Number)	Aadhaar OTP fallback
Status Tracking	ACTIVE (Immediate confirmation)	Webhook notification updates Postgres DB	Instant email & SMS alert on bounce

CORE MODULE: Relational Architecture, Data Integrity & Foreign Key Topology

Drizzle ORM schema mapping PostgreSQL 16 tables with UUID keys, timestamps, and ACID transactional consistency.

Table Name	Primary Key	Foreign Key Dependencies	Critical Attributes & Column Types
users	id (uuid)	None (Root identity entity)	email, phone_number, role (CLIENT, PARTNER, ADMIN), is_active
partners	id (uuid)	user_id -> users(id)	arn_number (348996), firm_name (Velocity Wealth), total_aum, commission
clients	id (uuid)	user_id -> users(id), partner_id	pan, aadhaar_last4, kyc_status, bank_account, digital_signature
instruments	code (varchar)	None (AMFI Master catalog)	name, fund_house, category, current_price (NAV), historical_data (JSONB)
sips	id (uuid)	client_id -> clients, instrument_code	amount, frequency, sip_day (1..28), status (ACTIVE, PAUSED), next_date
mandates	id (uuid)	client_id -> clients	umrn, max_amount, bank_name, status (ACTIVE), auth_mode (NET_BANKING)
transactions	id (uuid)	client_id -> clients, instrument_code	type (BUY, SELL, SIP_DEBIT), units, purchase_price, total_amount, status